

Working with Arrays

Objectives

- What are arrays
- Stages to creating arrays
- Arrays of primitives
- Arrays of object references
- Iterating over an array using a Java 5 for loop
- Passing arrays as variable size parameter lists

Variable Types

- Remember, there are three basic types of variable in Java
 1. Primitives
 2. Object references
 3. **Arrays**

What are Arrays

- Arrays are objects
- To declare an array, use `[]` syntax either
 - `Account[] myAccounts` ✓
 - `Account myAccounts[]`
- When initialized, arrays are of a **fixed size** and **type**
 - `Account[] myAccounts = new Account[3];`
- Arrays always start at index 0
- Have a public instance variable called **length**

Stages to Creating Arrays

1. Declaring
2. Creating
3. Populating

Arrays of Primitives

- When the array is created, the elements will have the default values 0, 0.0, or false

```
int[] myInts;           // declare the array
myInts = new int[5];     // create the array with 0's as elements

for (int i=0; i<myInts.length; i++) { // use the length property
    myInts[i] = i+1;       // place the values 1-5 in the array
}
```

Arrays of Object References

- When the arrays are created, the contents of the array are all null

```
Account[] myAccounts;           // declare the array
myAccounts = new Account[5];    // create the array (null
                                // references in the array)

for (int i=0; i<myAccounts.length; i++) {
    myAccounts[i] = new Account(); // create the array contents
}
```

Array Initializers

- Arrays can be created in one statement using **array initializers**

```
int[] myInts = { 4, 6, 7, 3, 7, 2 };           // note the curly brackets
String[] names = {"Rod", "Jane", "Freddy"};    // an array of strings
Account[] myAccounts = {
    new Account(),
    new Account(),
    new Account()
};                                              // an array of accounts
```

Iterating over an Array

- You can either use
 - A traditional 'for' loop
 - A 'for each' loop

for loop

```
for (int i=0; i<myAccounts.length; i++) {  
    myAccounts[i].addInterest();  
}
```

for each loop

```
for (Account a:myAccounts) {  
    a.addInterest();  
}
```

Variable Argument Lists

- Java methods take a fixed set of arguments
- Since Java 5, you can also use an array to make it easier to handle a variable argument list
 - Either pass in a variable number of individual items or
 - Pass in an array

```
public class VarargsTest {  
    public static void main(String[] args) {  
        String[] names = {"rod", "jane", "freddie"};  
        processArgs(names);  
        processArgs("Rod", "Jane", "Freddy", "Zippy");  
    }  
    public static void processArgs(String ... args) {  
        for (String text:args) {  
            System.out.println(text);  
        }  
    }  
}
```

Summary

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